ASSIGNMENT 3: TRAIN YOUR OWN LLMS - PARAMETER-EFFICIENT FINE-TUNING OF QWEN2-7B FOR FINANCIAL QUESTION ANSWERING

Hng Cherng Khai 125400013
Chinese University of Hong Kong
Shenzhen, China
125400013@link.cuhk.edu.cn

ABSTRACT

Large Language Models (LLMs) have demonstrated remarkable capabilities in general natural language tasks. However, their performance often degrades in specialized domains such as finance due to a lack of domain-specific training. Furthermore, fine-tuning these massive models requires substantial computational resources that are often prohibitive for individual researchers. This report investigates the efficacy of Parameter-Efficient Fine-Tuning (PEFT) using Low-Rank Adaptation (LoRA) to adapt the Qwen2-7B model for financial question-answering. Utilizing the Unsloth framework for 4-bit quantization, the model was fine-tuned on the finance-alpaca dataset. The experimental results demonstrate a significant improvement in model performance, with the evaluation perplexity dropping from a baseline of 6.97 to 3.33 after only 60 training steps. This study confirms that high-performance domain adaptation can be achieved with minimal computational overhead, offering a viable pathway for deploying specialized LLMs in resource-constrained environments.

1 INTRODUCTION

What is your research topic This research focuses on the efficient domain adaptation of Large Language Models (LLMs) for the financial sector. Specifically, the study implements **Supervised Fine-Tuning (SFT)** via **Low-Rank Adaptation (LoRA)**, which is a Parameter-Efficient Fine-Tuning (PEFT) technique to specialize the Qwen2-7B model. By utilizing the `finance-alpaca` dataset, the research aims to transform a general-purpose foundation model into a specialized assistant capable of adhering to specific instructions and generating accurate financial advice and definitions.

Why the task is important While foundation models like Qwen or Llama are highly capable, they often lack the nuanced vocabulary and context required for specialized fields like finance. Furthermore, traditional "full fine-tuning" of 7-billion parameter models requires massive computational resources. This task is crucial because it allows highly capable foundation models to be usable in specialized high-stakes domains like finance while drastically reducing the computing cost.

What you did and what you achieved To address this challenge, a training pipeline using the Unsloth optimization library has been utilized to fine-tune the Qwen2-7B model with 4-bit quantization. The model was trained on the `gbharti/finance-alpaca` dataset using LoRA adapters to update only a fraction of the model's parameters. As a result, the experiment yielded significant quantitative improvements in model's perplexity score which is a standard measure of how well a model predicts text. By training for just 60 steps, the perplexity score dropped from a baseline of 6.97 to 3.33. This represents a 52% improvement in the model's ability to process financial text, demonstrating that low-resource fine-tuning is highly effective for domain adaptation.

2 EXPERIMENT DESIGN

Definition of the task The core task addressed in this research is to implement Supervised Fine-Tuning (SFT) to adapt a general-purpose Large Language Model to the specific linguistic and conceptual patterns of the finance domain. The objective is to train the model to map financial instructions like "Explain the difference between ETFs and Mutual Funds" to accurate, context-aware responses. Unlike standard text completion, this task requires the model to adhere to an instruction-input-response format, minimizing the likelihood of hallucination while maximizing the coherence of complex financial explanations.

How do you design the entire experiment? The experiment is designed as a comparative study between a frozen pre-trained baseline and a parameter-efficient fine-tuned model. The pipeline consists of five distinct stages:

1. **Model Initialization:** The `unsloth/Qwen2-7B` model is loaded using 4-bit quantization to optimize memory usage for a single Tesla T4 GPU environment.
2. **Adapter Configuration:** Instead of full-parameter tuning, Low-Rank Adaptation (LoRA) is employed. Trainable rank decomposition matrices ($r = 16$, $\alpha = 16$) are injected into the model's attention ('q_proj', 'k_proj', 'v_proj', 'o_proj') and feed-forward layers ('gate_proj', 'up_proj', 'down_proj'), while the original weights remain frozen.
3. **Data Processing:** The `finance-alpaca` dataset is pre-processed into a standardized prompt structure and split into training (95%) and evaluation (5%) sets.
4. **Training Protocol:** The model is trained for 60 steps using the `SFTTrainer`, utilizing the AdamW 8-bit optimizer and a learning rate of 2×10^{-4}.
5. **Evaluation:** Finally, the experiment evaluates performance by calculating the perplexity metric on the held-out test set and conducting a qualitative study by comparing inference results with the LoRA adapters enabled versus disabled.

3 EXPERIMENTS

In this section, we evaluate the efficacy of the parameter-efficient fine-tuning (PEFT) approach. We compare the performance of the base Qwen2-7B model against the LoRA-adapted version across both objective metrics and subjective generation quality. The primary goal is to determine if low-rank adaptation can successfully instill domain-specific knowledge without compromising the model's linguistic stability.

3.1 EVALUATION SETUP

The experimental evaluation was conducted on a held-out test split derived from the `finance-alpaca` dataset. From the initial subset of 10,000 samples, 5% were reserved exclusively for validation to prevent data leakage. The evaluation was performed on a single Tesla T4 GPU utilizing the Unsloth optimization framework Unsloth AI (n.d.) to maximize throughput.

In this research, **Perplexity (PPL)** has been used as the primary quantitative metric. PPL measures the model's uncertainty in predicting the next token. A lower score of PPL indicates that the model is more confident and accurate in generating financial text. It is calculated as the exponent of the cross-entropy loss:

$$PPL = e^{\text{Cross Entropy Loss}}$$

3.2 QUANTITATIVE RESULTS

The training process demonstrated rapid convergence, with the training loss decreasing from 2.00 at step 1 to 0.98 at step 60. To assess the final model performance, we compared the evaluation loss and perplexity of the fine-tuned model against the baseline (pre-trained) model.

As shown in Table 1, the application of LoRA resulted in a dramatic performance improvement. The perplexity dropped by approximately **52%** (from 6.97 to 3.33). This significant reduction confirms

Model Configuration	**Evaluation Loss**	**Perplexity**
Baseline (Qwen2-7B)	1.9423	6.9748
Fine-Tuned (LoRA)	**1.2025**	**3.3283**

Table 1: Comparison of validation metrics between Baseline and Fine-Tuned models.

that even a short training run (60 steps) is sufficient for the model to adapt to the distribution of the financial dataset.

3.3 Case Study: Qualitative Comparison

To understand the practical impact of Low-Rank Adaptation (LoRA) beyond statistical metrics, we conducted a side-by-side comparison of the Baseline (Pre-trained) model and the Fine-Tuned model. For this inference, the temperature was set to 0 to ensure deterministic, reproducible outputs and to minimize the risk of hallucination.

3.3.1 Case 1: Definition and Concept Explanation

Query: ”What is the difference between a fixed-rate mortgage and an adjustable-rate mortgage?”

- **Baseline Output:** Provided a three-paragraph explanation. It defined both terms, explicitly listed the ”main advantage” of each, and offered a concluding summary on how to choose between them based on financial goals.
- **Fine-Tuned Output:** ”A fixed-rate mortgage is a loan with a fixed interest rate that remains the same throughout the life of the loan. An adjustable-rate mortgage (ARM) is a loan with an interest rate that can change over time, usually based on a benchmark interest rate. The interest rate on an ARM can increase or decrease depending on the market conditions, which can result in higher or lower monthly payments.”

Analysis: In this instance, the fine-tuning successfully induced *style transfer*. The `finance-alpaca` dataset typically consists of concise instruction-response pairs. The fine-tuned model adopted this distribution, stripping away the conversational fluff and ”pros/cons” analysis found in the baseline to provide a direct, dictionary-style definition. This represents a success in terms of adapting to the target data format.

3.3.2 Case 2: Complex Advisory Queries (The ”Knowledge Trade-off”)

Query: ”I have $10,000 and a long-term horizon. How should I think about diversifying my investments?”

- **Baseline Output:** Demonstrated broad general knowledge, suggesting specific asset classes including ”stocks, bonds, real estate, and commodities.” It also recommended specific vehicles like ”REITs” and ”index funds,” and mentioned the concept of rebalancing.
- **Fine-Tuned Output:** ”I would recommend investing in a mix of stocks and bonds. You can invest in a mix of stocks and bonds to diversify your portfolio. You can also invest in a mix of different types of stocks and bonds to further diversify your portfolio.”

Analysis: This result highlights a critical limitation known as *catastrophic forgetting* or knowledge degradation. While the Baseline model drew upon its vast pre-training to offer a nuanced strategy, the Fine-Tuned model collapsed into a repetitive loop (”mix of stocks and bonds”). Despite the low perplexity score (3.33) indicating high statistical confidence, the semantic quality of the generation degraded for open-ended advice. This suggests that while the model learned the *vocabulary* of finance, the short training duration (60 steps) or the limited diversity of the small dataset restricted its ability to formulate complex, multi-step advice.

3.3.3 Discussion

The comparison reveals a distinct trade-off. The Baseline model acts as a generalist tutor. It is verbose and knowledgeable but prone to generic responses. The Fine-Tuned model acts as a specialized glossary, delivering concise and direct response. However, the repetitive responses observed in Case 2 indicate a limitation in the Fine-Tuned model' s ability to handle tasks that require the integration and synthesis of multiple financial concepts (e.g., combining REITs, commodities, and rebalancing strategies). This suggests that the current LoRA adapter configuration may not sufficiently preserve the model' s original reasoning capabilities.

4 Conclusion

This research successfully demonstrated the feasibility and efficacy of fine-tuning the Qwen2-7B Large Language Model for the financial domain using Low-Rank Adaptation (LoRA) and 4-bit quantization. By training on the `finance-alpaca` dataset for just 60 steps on a single T4 GPU, we achieved a **52% reduction in perplexity** (from 6.97 to 3.33), confirming that parameter-efficient methods can rapidly adapt a model's probability distribution to a new domain with minimal computational resources.

Beyond the quantitative improvements, the qualitative evaluation offered important insights into how the model' s behavior changed through fine-tuning. The fine-tuned model adopted the concise and definition-driven style characteristic of the finance-alpaca dataset, showing that LoRA is highly effective for stylistic alignment. This highlights LoRA' s strength in teaching the model how to communicate in a domain-specific manner, not just what information to provide.

However, the experiments also revealed the limitations of low-resource tuning. The repetitive responses observed in the multi-concept diversification query underscore a trade-off, that is while domain-specific fluency improved, the model' s broader reasoning ability weakened. This suggests that the current training configuration and method may be sufficient for vocabulary and style adaptation but insufficient for preserving deeper, multi-step reasoning capabilities.

In summary, this project confirms that LoRA is a practical and accessible approach for developing specialized LLMs, especially under constrained computational budgets. Future work should explore adjusting the LoRA rank, increasing training steps, and incorporating more diverse financial reasoning tasks to achieve a more balanced integration of domain specialization and general reasoning performance.

ACKNOWLEDGMENT

This is the Assignment3 for CSC6052 / MDS5110 / CSC5051, see details in `https://nlp-course-cuhksz.github.io/`.

REFERENCES

Unsloth AI. Unsloth: Open source fine-tuning & rl for llms. `https://unsloth.ai/`, n.d. Accessed: 2025-11-24.